

AARAV CHARAK

Computer Science Engineering – Artificial Intelligence & Machine Learning

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Profile

Computer Science Engineering student specializing in Artificial Intelligence and Machine Learning, with hands-on experience in Python, C, embedded systems, and hardware prototyping. Interested in building practical systems that combine software, electronics, and intelligent technologies. Strong interest in research, problem solving, robotics, artificial intelligence, and developing unconventional technology projects.

Education

Lovely Professional University	2025 – 2029
B.Tech. in Computer Science and Engineering – Artificial Intelligence & Machine Learning	Phagwara, Punjab
KC International School	Graduated 2024
Class XII	Jammu, Jammu & Kashmir

Technical Skills

Programming: Python, C, C++ (Learning)

Electronics & Embedded Systems: ESP32, Arduino, GPIO, TFT Displays, Hardware Prototyping, Microcontroller Communication

Professional Skills: Analytical Problem Solving, Critical Thinking, Teamwork, Communication, Collaborative Decision-Making, Adaptability

Problem-Solving Approach: Carefully analyze problems before developing solutions, consider multiple possible approaches, and use team feedback to improve decisions and implement practical solutions collaboratively.

Interests: Robotics, Embedded Systems, Cybersecurity, Computer Hardware, Research & Experimental Projects

Projects

Echo Pod – DIY Portable Music Player	2026
<i>Raspberry Pi, Python, GPIO, USB-C, Embedded Systems</i>	

- Designed the architecture for a custom portable music player inspired by classic dedicated music players.
- Explored GPIO-based physical controls, rotary-wheel input, display integration, storage, power management, and USB-C audio.
- Investigated hardware/software integration and modular system architecture for building a fully custom portable media device.

Smart Switch Board	Jan 2026 – May 2026
<i>ESP32, Arduino, TFT Display, Embedded Systems</i>	

- Developed a smart switch board capable of controlling household lights and fans through an ESP32 and Arduino-based system.
- Implemented fan-speed control and appliance switching through a TFT touchscreen interface.
- Designed communication between microcontrollers and integrated hardware control with a graphical user interface.
- Gained practical experience in embedded programming, electronics, microcontroller communication, and hardware prototyping.

Achievements

- Solved **100+** programming and coding problems across programming practice platforms.
- Achieved **90+** coding questions in Python and SQL through CodeTantra practice.
- Achieved **90+** coding questions in C through structured programming practice.

Certifications

- **Programming for Everybody (Getting Started with Python)** – University of Michigan, Coursera [Certificate]
- **Python Data Structures** – University of Michigan, Coursera [Certificate]
- **Using Python to Access Web Data** – University of Michigan, Coursera [Certificate]

Research & Technical Interests

Artificial Intelligence		Machine Learning		Robotics		Embedded Systems
Cybersecurity		Electronics		Computer Hardware		Experimental Technology